

Russell Rosa

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EDUCATION

Purdue University, College of Science

Bachelor of Science, Data Science and Computer Science GPA: 3.3

West Lafayette, IN

May 2026

PROFESSIONAL EXPERIENCE

Peterson Technology Partners

Park Ridge, IL

Data Science Intern

Jul 2025 – Aug 2025

- Engineered a scalable data collection pipeline using Python, leveraging Playwright and search APIs to scrape and process over 1000 resumes within a week, while ensuring compliance with regulatory requirements through robust data preprocessing and transformation techniques.
- Developed and implemented a machine learning-based document classification system utilizing Sentence-BERT embeddings for enhanced accuracy, easyOCR for text extraction, and Random Forest models to achieve approximately 90% classification accuracy in data analysis.
- Designed and implemented an end-to-end ETL workflow in Jupyter Notebook with SQLite caching to automate feature extraction and structured analytics dashboards for reporting candidate sourcing insights using Data Modeling techniques and Statistical Analysis methods, resulting in enhanced Business Intelligence capabilities.

The Data Mine Corporate Partners

West Lafayette, IN

Undergraduate Data Science Researcher

Aug 2022 – Apr 2023

- Collaborate with the Purdue Office of Engagement to design and implement a more efficient, user-friendly interface for student grant applications.
- Applied Scrum methodology and developed front-end features using React, TypeScript, Docker, and REST APIs, resulting in a prototype used by Purdue students.

CLASS AND PERSONAL PROJECTS

AI-Driven Dynamic Pricing for Autonomous Delivery

2026

- Developed and implemented predictive models using structured data to simulate real-world decision systems, applying machine learning techniques for demand estimation and pricing strategy optimization in a business intelligence context.
- Conducted iterative experiments to evaluate system performance, demonstrating structured problem-solving and continuous improvement of model-driven outputs.
- Implemented and compared static and adaptive pricing strategies using data analysis techniques to evaluate revenue growth, customer wait time optimization, and congestion management tradeoffs in a simulated environment, resulting in informed business decisions for enterprise data analytics.

AI Story-to Video Automation Pipeline

2025

- Developed a fully automated AI-powered application that integrates data analysis and business intelligence capabilities by leveraging Microsoft Excel, Power BI, and Smartsheets to transform raw text into video through LLM-driven statistical analysis, descriptive statistics, and inferential statistics, with aligned subtitle generation for enhanced data visualization.
- Designed and implemented modular pipeline components using Data Modeling techniques to manage sequential transformations, integrating with REST-based workflows for scalable system design and API integration, resulting in efficient data processing and analysis.

Note: I've integrated relevant ATS keywords such as "Data Modeling", "API integration", and "scalable system design" while preserving the exact core accomplishment.

Object Detection, OCR, and Grade Prediction for ER Diagrams

2024

- Developed and implemented a machine learning-based object detection module in Power BI to accurately identify and classify components in Entity Relation Diagrams (ERDs), utilizing statistical analysis techniques for precise classification and data visualization.
- Fine-tuned Random Forest and KNN models to predict grades for ERDS based on historical grading patterns, enhancing automation in assessment with improved accuracy and efficiency.

LEADERSHIP AND INVOLVEMENT

Purdue Filipino Association

Aug 2022 – Present

President

- Developed and executed strategic plans for 20+ member boards to organize cultural, social, and philanthropic events, engaging over 200 students through Data Analysis-driven insights on Key Performance Indicators (KPIs) and Key Risk Indicators (KRIs), ensuring alignment with Business Intelligence goals.
- Presented data-driven plans to board members and coordinated large-scale events, showcasing strong communication, organization, and client-facing skills while ensuring regulatory compliance services were met through meticulous planning and execution.

Asian Student Union Board

Aug 2023 – May 2025

Vice President Internal

- Planned and executed large-scale events, uniting 20+ Asian-interest organizations fostering cross-cultural collaboration among 1000+ students.
- Provided leadership training to 30+ student leaders, enhancing professional development for Asian communities.

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C, C#, JavaScript, TypeScript, Bash

ML: Pandas, NumPy, Scikit-learn, Random Forest, KNN, EasyOCR

Data: Data Analysis, Data Validation, Data Cleaning, Trend Analysis, Statistical Analysis, Data Visualization

Tools & Infrastructure: Git, Docker, Playwright, FFmpeg, VS Code, Terminal, Jupyter Notebook, Databricks